

CPT304 Software Engineering 2

Coursework 1 – Research-Led Software Enhancement

Start date	7 th April 2026
Submission date	11 th May 2026
Assignment type	Group Work – 4 members
Percentage in final marks	40% of Total Module Mark (Pass Mandatory)
Late submission policy	5% of the awarded marks shall be deducted for each working day after the submission date, up to a maximum of five working days. Late submission for more than five working days will be not accepted.
Important notes	• Plagiarism results in award of ZERO mark; all submissions will be checked by TurnItIn. • AI Permissions: Permitted with restrictions: Students may use AI solely as a language assistant. • The formal procedure for submitting coursework at XJTLU is strictly followed. Submission link on Learning Mall will be provided in due course. The submission timestamp on Learning Mall will be used to check late submission.

Learning Outcome

- D. Present, analyse, and give a reasoned critique of articles in the software engineering research literature.
- E. Apply an integrated or systems approach to the deployment of software systems that meet business goals, considering applicable safety, diversity, inclusion, cultural, and societal issues.

1. The Mission

Your team will take a "tiny" open-source project and elevate it to professional, responsible standards. You must prove your ability to **audit** a system, **research** solutions in software literature, and **deploy** a compliant application.

2. Submission Deliverables

Submit one ZIP file named *GroupXX.zip* containing:

- *report.pdf* (Maximum 1,500-word Report).
- *github-url.txt* (Link to your team's forked repository).
- *live-url.txt* (Link to the production-deployed app).

- *individual-contribution.xlsx* (Peer-assessment marks).

3. The Workflow

1. **Project Selection:** Your group should discuss and decide one project to work on. Fork **one** repository from the Project List listed in section 6 below.
2. **Audit Source Code:** Use tools (axe DevTools, Lighthouse, manual inspection, and others) to identify **4 specific flaws (Deficiencies)** in the original code.
 - *Note: These must be unique problems found in the source code, not the missing requirements listed in Step 4.*
3. **Research:** Find 4 external articles (Academic papers from IEEE/arXiv or professional blogs like dev.to/Medium) that provide a methodology for fixing these 4 specific flaws.
4. **Refactor The Project:** Implement your researched fixes **AND** ensure the app meets the following **5 Minimum Baseline Standards**:
 - **Live Uptime:** Site must be live on Vercel/Render for 7+ consecutive days.
 - **Test Coverage:** Minimum **80%** (proven via Codecov/Istanbul badge).
 - **Accessibility:** Lighthouse Accessibility score of **90+**.
 - **Internationalization (i18n):** Working toggle for at least two languages.
 - **Legal Compliance:** A functional Cookie Banner and a dedicated Privacy Policy page.
5. **Documentation:** Use the provided template to write your report with maximum 1,500 words.
6. **Assessment:** Complete the individual contribution form.

4. Report Template (Mandatory Structure)

Max: 1,500 Words ($\pm 10\%$)

- **Section 1: App Selection** 100 words (Which app and why?)
- **Section 2: Deficiency Analysis** 300 words
 - *Section 2.1: Detection:* How did you identify the flaw? (e.g., "axe DevTools flagged X"). 60/300 words (Recommendation)
 - *Section 2.2: Literature:* Summarize the article/paper used to find a solution. 100/300 words (Recommendation)
 - *Section 2.3: Implementation:* 140/300 words (Recommendation)
 - How your research led to the implementation.
 - Show a "Before vs. After" code snippet.

(Repeat this section for each deficiency i.e. Section 3, 4, 5 for 3 other deficiencies)

- **Section 6: Baseline Standards Evidence** 100 words
 - Screenshots of: Vercel 7-day log, Codecov 80% badge, Lighthouse 90+ score, Language switch, and Cookie banner.
- **Section 7: Individual Contribution Forms** Word count are excluded
 - Contribution forms of all 4 members.
- **Section 8: References:**

- Minimum 4 citations (IEEE, arXiv, or high-quality dev blogs).
- Students must use the IEEE Citation Style for all technical literature.

Additional Content and Formatting Requirement

- **Implementation Rationale:** The Implementation section of each deficiency must explicitly explain how the research directly informed your code changes; simply stating "I followed the guide..." is insufficient for high marks.
- **Visual Evidence:** Every screenshot (e.g., Vercel 7-day log, Codecov badge, Lighthouse score) must include a figure number and a descriptive caption.
- **Code Snippets:** Use a monospaced font (e.g., Consolas) for "Before vs. After" code blocks. Note that standard code is typically excluded from word counts, but the explanatory text surrounding it is included. Code snippets must be concise enough for easy understanding.
- **Typography and Font Requirements**
 - Body Text: Use 12pt Calibri or Arial for readability.
 - Headings:
 - Level 1 Headings (Section Titles): 14pt, Bold.
 - Level 2 Headings (Sub-sections): 12pt, Bold.
 - Code Snippets: Use a monospaced font like 10pt Consolas or Courier New to clearly distinguish "Old vs. New" code.

5. Marking Scheme

Research Quality (50 Marks)

- **High-Standard (40–50 Marks):** Precise detection with tools; high-tier literature cited; includes clear **Old vs. New** snippets; and a sophisticated **Implementation Rationale**.
- **Standard (20–39 Marks):** Logical fixes applied but may rely on simpler sources (e.g., dev blogs) or have a weaker "Logic Bridge" between research and code.
- **Weak (0–19 Marks):** Superficial flaws; missing code snippets; or no clear connection between research and fix.

Baseline Standards (30 Marks)

- Marks are awarded based on **Visual Evidence** in Section 6 of the report.
- Every screenshot must include a figure number and a descriptive caption to be considered valid evidence.

Individual Contribution (20 Marks)

- The marker must cross-reference the **Individual Contribution Form** data with GitHub commit percentages and specific tasks.
- Students are required to provide direct links to their **Pull Requests Authored** for verification.

Formatting Compliance Check

Submissions that fail to meet these mandatory formatting requirements should be penalized under the "Standard" or "Weak" categories:

- **Typography:** 12pt Calibri/Arial for body; 14pt Bold for Level 1 Headings.
- **Code Blocks:** Must use 10pt monospaced fonts (Consolas/Courier New).
- **Word Count:** 1,500 words ($\pm 10\%$), excluding code.

6. Project List

Project Name	GitHub Repository URL
Advance Finance Tracker	https://github.com/sptin2002/advanced-finance-tracker
Personal Budget App	https://github.com/sptin2002/Budget-app
Inventory Management System (UNIQLO)	https://github.com/sptin2002/inventory-management-system
Inventory Management System	https://github.com/sptin2002/Inventory-App-JS
Order Management	https://github.com/sptin2002/biztrack
HR System	https://github.com/sptin2002/Employee-Management-System
E-Commerce	https://github.com/sptin2002/E-Commerce
Seat Booking App	https://github.com/sptin2002/seat-booking-app
Task Management App	https://github.com/sptin2002/Taskify
Interactive polling and presentation software	https://github.com/sptin2002/polln

7. Support Schedule:

- **Week 7 Lab:** System Deployment Workshop.
- **Consultation Hours:** Visit the Module Leader and Teaching Assistants for coding advices.